TEXTURE UNIT, TEXTURE SPECTRUM AND TEXTURE ANALYSIS

[1]Dong-Chen He and [2]Li Wang

[1]Université de Sherbrooke
Sherbrooke, Québec, Canada J1K 2R1

[2]Canada Centre for Remote Sensing
1547 Merivale Road, Nepean, Ontario, Canada K2G 3J4

ABSTRACT

This paper describes a new statistical method for texture analysis, based on the concept of a *Texture Unit*. The textural information of an image can be revealed by its *Texture Spectrum*. A preliminary evaluation study demonstrates the potential usefulness of the proposed method for texture analysis in remote sensing.

Keywords: Texture Analysis, Texture Unit, Texture Spectrum, Pattern Recognition.

1. INTRODUCTION

As the spatial resolution of satellite image data increases (80 meters for Landsat-MSS, 30 meters for Landsat-TM and 10 meters for Spot-PLA), texture analysis plays a more important role in image processing, image classification, and in interpretation of remotely sensed data. This means that image classification and interpretation based on the analysis of individual pixels will be no longer sufficient to satisfy the needs of landscape mapping and cartography. A good understanding or a more satisfactory interpretation of remotely sensed imagery should include the description of both spectral and textural aspects.

According to Haralick (1979), methods of texture analysis are usually divided into two major categories. The first is the *structural approach*, where texture is considered as repetition of some primitives with a certain rule of placement. The traditional Fourier spectrum analysis is often used to determine the primitives and the placement rule. Several authors have applied this method in texture classification and texture characterization, with a certain degree of success (He et al., 1988, D'Astous and Jernigan, 1984, Matsuyama et al., 1980). Problems may be encountered in practice to identify the primitives and the placement rule in natural images such as some remotely sensed data.

The second major approach in texture analysis is the *statistical method*. Its aim is to characterize the stochastic properties of the spatial distribution of grey levels in an image. The grey tone co-occurrence matrix is frequently used for such characteristics. A set of textural features derived from the co-occurrence matrix has been widely used in practice to extract textural information from digital images (He et al., 1987, Conners, 1979, Haralick, 1973). Sometimes, this kind of second order grey level co-occurrence matrix produces unsatisfactory results. Some reasons for this are as follows. Firstly, the matrix depends not only on the spatial relationships of grey levels but also on regional intensity background variation within the image. Secondly, the co-occurrence matrix reveals textural information of the image in a given displacement vector $\vec{v} = (\Delta x, \Delta y)$, so that the choice of this vector is somewhat problematic.

The purpose of this paper is to present a new statistical method of texture analysis focused on texture characterization and discrimination. The concept of *Texture Unit* is proposed first. It may be considered as the smallest complete unit which best characterizes the local texture aspect of a given pixel in all the eight directions of a square raster. Then, a texture image is characterized by its *Texture Spectrum*, which describes the distribution of all the Texture Units within the image. Some natural images have been used to evaluate the discriminating performance of the Texture Spectrum. The results lobtained here demonstrate the potential usefulness of the proposed method in texture analysis. These methods have been more extensively discussed elsewhere by He and Wang (1989a,b) and Wang and He (1989).

2. METHODOLOGY

2.1 Texture Units

In a square raster digital image, each pixel is surrounded by eight neighbouring pixels. The local texture information for a pixel can be extracted from a neighbourhood of 3x3 pixels, which represents the smallest complete unit (in the sense of having eight directions surrounding the pixel).

Given a neighbourhood of 3x3 pixels, which will be denoted by a set containing nine elements: $V = \{V_o, V_1, \ldots V_8,\}$, where V_o represents the intensity value of the central pixel and $V_i, (i = 1,2,\ldots 8)$, is the intensity value of the neighbouring pixel i, we define the corresponding *Texture Unit* by a set containing eight elements, $TU = \{E_1, E_2, \ldots, E_8\}$, where $E_i, (i = 1,2,\ldots 8)$ is determined by the formula:

$$E_i = \begin{cases} 0 \text{ if } V_i < V_o \\ 1 \text{ if } V_i = V_o; \\ 2 \text{ if } V_i > V_o \end{cases} \quad \text{for: } i=1,2,\ldots,8.$$

and the element E_i occupies the same position as the pixel i.

Authorized licensed use limited to: UNIV ESTADUAL PAULISTA JULIO DE MESQUITA FILHO. Downloaded on October 16,2026 at 00:26:12 UTC from IEEE Xplore. Restrictions apply.

As each element of TU has one of three possible values, the combination of all the eight elements results in $3^8 = 6561$ possible Texture Units in total.

2.2 Labeling Texture Units

There is no unique way to label and order the 6561 Texture Units. In our study, the 6561 Texture Units (N_{TU}) are labeled by using the following formula:

$$N_{TU} = \sum_{i=1}^{8} 3^{i-1} E_i, \qquad N_{TU} \in \{0,1,2,\ldots 6560\}$$

where E_i is the ith element of Texture Unit set TU = $\{E_1, E_2, \ldots, E_8\}$.

In addition, the eight elements may be ordered differently. If the eight elements are ordered clockwise as shown in Figure 1, the first element may take eight possible positions from top-left (a) to middle-left (h), then the 6561 Texture Units can be labelled by the above formula under eight different ordering ways (from a to h).

Figure 2 gives an example of transforming a neighbourhood to Texture Unit with the Texture Unit number under the ordering method a.

2.3 Texture Spectrum

The previously defined set of 6561 Texture Units describes the local texture aspect of a given pixel, that is, the relative grey level relationships between the central pixel and its eight neighbours. Thus, the statistics on frequency of occurrence of all the Texture Units over a whole image should reveal texture information. We termed *Texture Spectrum* the frequency function of all the Texture Units, with the abscissa indicating Texture Unit number N_{TU} and the ordinate representing its occurrence frequency.

In practice, a real texture image is usually composed of two parts: texture elements and random noise or background. The greater the proportion of texture components compared to the background, the better that texture can be perceived by human vision. In the Texture Spectrum, the increase in percentage of texture components in an image will result in a tendency to form a particular distribution of peaks. In addition, different textures are composed of particular Texture Units with different distributions in their Texture Spectra. In this way, the texture of an image can be characterized by its Texture Spectrum.

It should be noted that the labeling method chosen may affect the relative positions of Texture Units in the Texture Spectrum, but will not change their frequency values in the Texture Spectrum.

3. EVALUATION

In order to evaluate the performance of Texture Spectrum in texture characterization and classification, several experimental studies have been carried out on four of Brodatz' natural texture images (Brodatz, 1966). These images were selected because they are broadly similar to one another, and also they resemble parts of remotely sensed images. Each image shown in Figure 3 consists of 256x256 pixels with 64 normalized grey levels.

3.1 Texture Spectra

If the Texture Spectrum has discriminating performance for different textures, different texture images should have correspondingly different spectra. This leads up to calculate the Texture Spectra for the four images of Figure 3 with eight ways of ordering (from a to h). The results are illustrated in Figure 4. We note that different textures have correspondingly different spectra when comparing the number, position, width and height of their principal peaks, demonstrating the clear discriminating performance of the Texture Spectrum.

3.2 Texture Classification

In addition to the visual evaluation over the Texture Spectra, a further quantitative study was performed using a supervised classification over the four texture images of Figure 3. A sample subimage of 30x30 pixels was randomly selected over each texture. Using a window of 30x30 pixels together with a step of two pixels in the row and column directives, the full image of Figure 3 has been processed and each central pixel of the window was assigned to one of the four texture classes. The minimum distance decision rule was employed and the absolute difference between two Texture Spectra was taken as the distance between them. The result is illustrated in Figure 5, representing 98.5% correct classification for texture image A; 96.4% for texture B; 98.9% for Texture C and 95.9% for texture D, resulting in an average classification accuracy of 97.5%.

It should be noted that the above correct classification rates are calculated over all the pixels, including the regions near the boundaries of the four textures. If we remove these pixels from the counter, the correct classification rate will be respectively 100% for textures A, B and C, and 98.4% for D, representing an average recognition rate of 99.6%.

It should also be noted, that the bin pattern for the texture spectrum, shown in Figure 4, represents 6561 possible texture units of which only part will be statistically meaningful to characterize a particular texture. A future extension of the method presented here, would have to extract the significant bins for each class and associate bins that represent slightly different patterns. This is equivalent to constructing a structure for the initially arbitrary encoding of so many bins. During classification, for a given window, the significant level of bin occupancy will need to be determined and compared with the corresponding bins in the training set.

4. DISCUSSION

Based on the concept of the *Texture Unit* and *Texture Spectrum*, a new statistical method of texture analysis has been presented. Preliminary evaluations show that the Texture Spectrum is able to reveal texture information in digital images and has promising discriminating performance for different textures. In addition, when compared with the other statistical methods, the proposed method possesses several advantages:

1. The Texture Unit method extracts the local texture information for a given pixel from a neighbourhood of 3x3 pixels, i.e., in all the eight directions from the central pixel instead of only computing one displacement vector as is done for the grey level

Authorized licensed use limited to: UNIV ESTADUAL PAULISTA JULIO DE MESQUITA FILHO. Downloaded on October 16,2026 at 00:26:12 UTC from IEEE Xplore. Restrictions apply.

co-occurrence matrix. So, in this respect, the new approach is more complete for the characterization of textural properties.

2. Since the definition of the set of 6561 Texture Units is independent of the images analyzed, we could expect to describe all the forms of texture in a unified way.

3. The textural aspect of an image is characterized in the form of a spectrum, making it possible to apply the Texture Spectrum concept to other problems of image processing, such as in designing digital filters using Texture Spectrum.

4. The proposed method can be easily adapted to texture or shape analysis of binary images. There are only two possible values (0 and 1) for each element of the Texture Unit. The total number of Texture Units becomes $2^8 = 256$. The statistics and calculation time will be reduced.

The methods described here are currently being applied to an integrated set of airborne SAR, SPOT, SEASAT and LANDSAT MSS images for mineral exploration described in Fabbri et al., (1989).

REFERENCES

Broadtz, P., 1966, Texture - A photographic album for artists and designers, Cover, New York.

Conners, R.W., 1979, "Toward a Set of Statistical Features which Measure Visually Perceivable Qualities of Textures", in Proc. Pattern Recognition Image Processing Conf., pp.382-390.

D'Astous, F. and M.E. Jernigan, 1984, "Texture Discrimination Based on Detailed Measures of Power Spectrum", Proc. IEEE Comput. Societ. Conf. on Pattern Recognition and Image Process., pp.83-86.

Haralick, R.M., K. Shanmugan and I. Dinstein, 1973, "Textural Features for Images Classification", IEEE Trans. Syst., Man, Cybern., Vol. SMC-8, Nov, pp.610-621.

Haralick, R.M., 1979, "Statistical and Structural Approaches to Texture", Proc. IEEE, Vol. 67, pp.786-804.

He, D.C. and Wang, L., 1989a, "A New Method for Texture Analysis - Texture Spectrum" (submitted to Pattern Recognition).

He, D.C. and Wang, L., 1989b, "Texture Filters Based on Texture Spectrum" (submitted to Pattern Recognition).

He, D.C., L. Wang and J. Guibert, 1987, "Texture Features Extraction", Pattern Recognition Letters, 6, pp.269-273.

He, D.C., L. Wang and J. Guibert, 1988, "Texture Discrimination Based on an Optimal Utilization of Texture Features", Pattern Recognition, No. 2, pp.141-146.

Matsuyama, T. et al., 1980, "Structural Analysis of Natural Textures by Fourier Transformation", Compt. Vision Graphics Images Process., 12, pp.286-308.

Fabbri, A.G., C.A. Kushigbor, L. Wang, A.B. Baker and S.M. Roscoe, 1989, "Integration of Remotely Sensed and Ancillary Geological Data for Assessing Mineral Resources in the Bathurst Inlet Area, N.W.T., Canada", Proc. 21st Application of Computers and Operations Research in the Mineral Industry, Feb. 27-March 2, 1989, Las Vegas, Nevada, Society of Mining Engieers Inc., p.1109-1120.

Wang, L. and He, D.C., 1989, "Texture Classification Using Texture Spectrum" (submitted to Pattern Recognition).

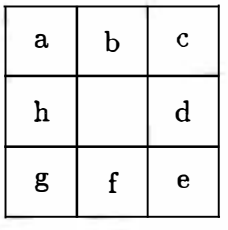


FIGURE 1. Eight clockwise successive ordering ways of the eight elements of texture units: the first element E_1 may take eight possible positions from a to h.

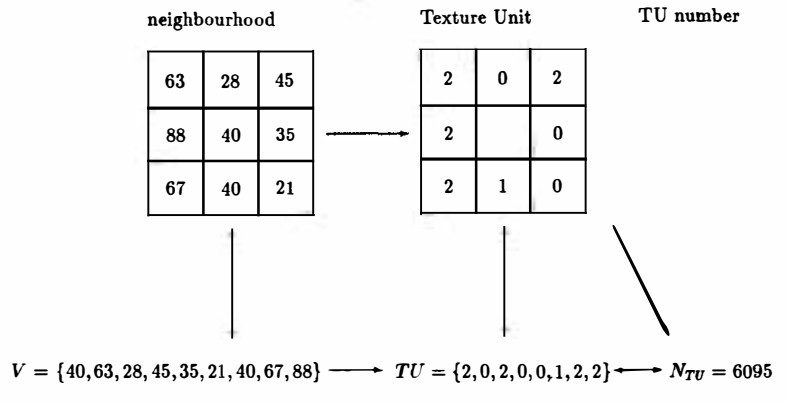


FIGURE 2. Example of transforming a neighbourhood to texture unit with texture unit number.

Authorized licensed use limited to: UNIV ESTADUAL PAULISTA JULIO DE MESQUITA FILHO. Downloaded on October 16,2026 at 00:26:12 UTC from IEEE Xplore. Restrictions apply.

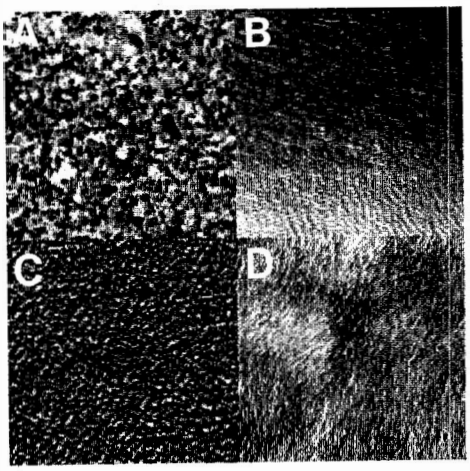


FIGURE 3. Four of Brodatz' texture images.

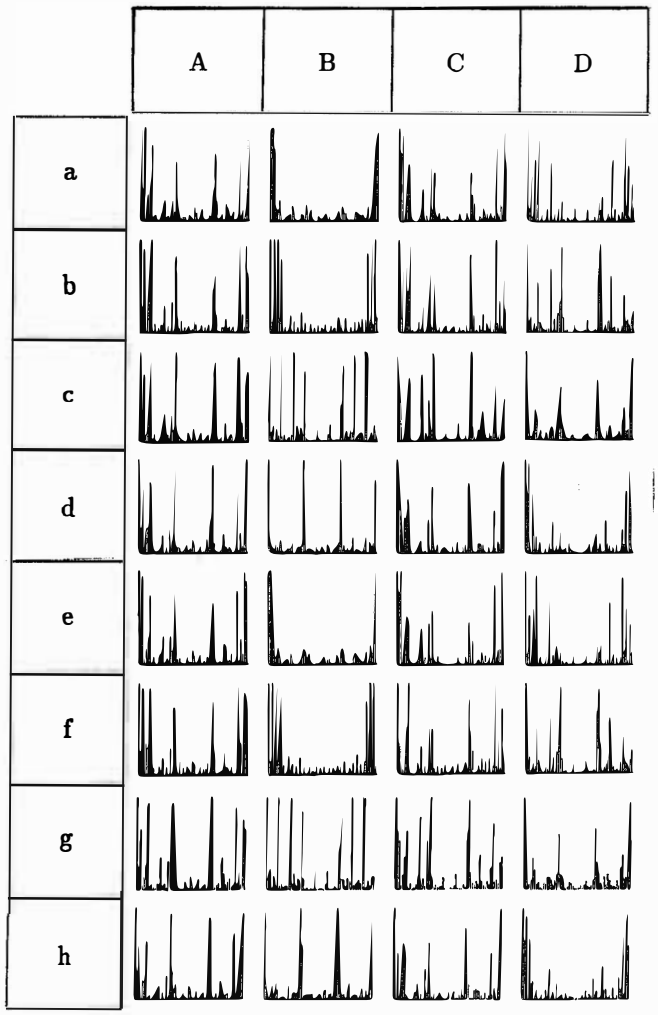


FIGURE 4. Texture Spectra of four images (from A to D) under eight ordering ways (from a to h).

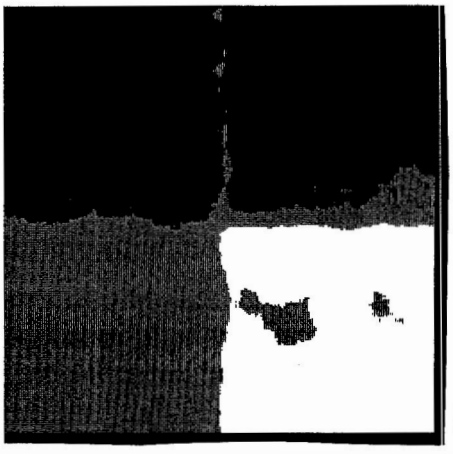


FIGURE 5. Classification of Figure 3.

Authorized licensed use limited to: UNIV ESTADUAL PAULISTA JULIO DE MESQUITA FILHO. Downloaded on October 16,2026 at 00:26:12 UTC from IEEE Xplore. Restrictions apply.